STATEMENT OF WORK

Neuropathology characterization for the Brain Atlas Project

GENERAL INFORMATION

Statement of Need and Purpose:

The National Institute on Aging (NIA) requires neuropathology services support of the Brain Atlas project which was established in 2022. NIA's Center for Alzheimer's Disease and Related Dementias is working to better understand the differences between healthy brains and brains affected by Alzheimer's disease at the cellular level. Researchers are working to create a human brain atlas or map of different brain cell types based on their gene activity. Researchers can use this atlas to investigate cell types present in brain regions affected in disease. Many different cells in a diseased brain experience change in their gene expression and downstream intracellular activities. The neuropathology analysis will lead to a better understanding of Alzheimer's Disease and the prioritization of potential target genes, pathways, and future therapeutics.

Background Information and Objective:

Contractor shall provide neuropathology services to assist in the creation of a human brain atlas or map of different brain cell types based on their gene activity.

Period of Performance:

June 1, 2026 – May 31, 2027

SCOPE OF WORK

General Requirements:

Independently and not as an agent of the Government, the Contractor shall furnish all the necessary services, qualified personnel, material, equipment, and facilities, not otherwise provided by the Government as needed to perform the Statement of Work below.

Specific Requirements:

The Contractor shall provide neuropathology services on 58 large, multiple piece brain samples as listed below:

- Trimming, cassette printing, processing and wet embedding into paraffin for 16 blocks for each sample

- Routine hematoxylin and eosin morphology staining of tissues for 16 blocks for each sample
- Chromogenic immunostaining on 30 slides per sample, using a biotin-free assay:
    - Phospho-TDP-42 clone 11-9, Cosmo Bio
    - Phospho-Ser129 alpha-synuclein, Abcam
    - Beta-amyloid, Abcam
    - Phospho-Tau (Ser202, Thr205) Monoclonal Antibody (AT8), Biotin, Invitrogen
- Digital 40x scanning and preservation of images for 12 months on contractor's platform. Platform should include:
    - A minimum of 67TB of storage on vendor's platform
    - 3 Users with ability to access to view, annotate, snapshot, and download raw files.
- Contractor shall provide all slides related to the 58 samples
- Contractor shall provide neuropathology services within 5-10 business days after receiving samples
- Contractor shall provide pick up and delivery from NIH main campus in Bethesda, Maryland

GOVERNMENT RESPONSIBILITIES

The government will provide 58 brain sample cases, and required antibodies.

DELIVERY OR DELIVERABLES

Contractor shall provide trimming, cassette printing, and wet embedding on 16 blocks for 58 brain samples. Contractor shall provide immunostaining on 30 slides for each of the 58 samples Contractor shall provide 40x scanning and image preservation for all images related to the 58 samples. Contractor shall provide platform for viewing, annotating, and downloading raw files. Contractor shall provide slides used for analysis.

REPORTING REQUIREMENTS

The contractor shall provide a one time report documenting samples processing procedures.

OTHER CONSIDERATIONS

Travel:

The principal place of performance is the contractor's work site

Key Personnel:

The contractor shall provide a dedicated project manager to the government as a point of contact for the services.

Lowest Price Technically Acceptable (LPTA) Evaluation Criteria

All requirements below are Pass/Fail under an LPTA (Lowest Price Technically Acceptable) procurement.

1. Technical Acceptability (Pass/Fail)

An Offeror will receive a Pass if the proposal clearly demonstrates the ability to meet all mandatory technical requirements of the SOW. Failure to meet any single requirement results in an automatic Fail.

1.1 Required Sample Processing Capabilities

- Offeror must demonstrate the capability to perform the following for 58 large, multi-piece brain samples:
- Trimming, cassette printing, processing, and paraffin embedding for 16 blocks per sample.
- Routine hematoxylin and eosin (H&E) staining for 16 blocks per sample
- Chromogenic immunostaining on 30 slides per sample, using all required antibodies (TDP-43, α-synuclein, beta-amyloid, AT8)

Pass criteria: Offeror fully demonstrates the capability to perform all tasks at required volumes with specified assays.

1.2 Digital Imaging & Data Platform Requirements

Offeror must provide:

- 40× digital slide scanning for all images
- 12-month preservation of all images on the contractor's platform

- A system that includes:
    - At least 67 TB of secure storage
    - 3 user accounts with permissions to view, annotate, snapshot, and download raw files

Pass criteria: All platform requirements are met without exceptions.

1.3 Schedule & Turnaround

Offeror must demonstrate ability to complete:

- All neuropathology services within 5–10 business days after receipt of samples
- Pickup and delivery service to/from NIH main campus, Bethesda, MD

Pass criteria: The proposal clearly meets both turnaround and logistics requirements.

1.4 Deliverables Compliance

Offeror must commit to providing:

- All processed blocks
- All stained slides
- All scanned images
- Digital platform access
- One-time report of sample processing procedures

Pass criteria: All SOW deliverables are addressed with no exceptions.

1.5 Staffing & Key Personnel

Offeror must identify:

- Adequately trained personnel for histology, immunostaining, and imaging
- A dedicated Project Manager, as required in the SOW

Pass criteria: All key roles and qualifications are clearly provided.

1.6 Facilities & Equipment

Offeror must certify access to:

- Appropriate histology laboratory facilities

- Equipment sufficient for required processing, staining, and high-resolution digital imaging
- Secure IT infrastructure for digital storage and user access

Pass criteria: Facilities and equipment are fully sufficient and described.

2. **Price Evaluation (Lowest Price Technically Acceptable)**
   - Only Offerors who receive a Pass on all technical factors will move to price evaluation.
   - Award will be made to the lowest-priced Offeror who meets all technical requirements.
   - Price will be reviewed for reasonableness and completeness.

Instructions to Offerors

Offerors shall submit proposals in accordance with the instructions below. Failure to submit all required information may result in the proposal being deemed Technically Unacceptable.

1 Proposal Format

- Page Limit: Typically 20–30 pages unless agency policy differs.
- Font: 11- or 12-point, single spaced.
- Sections Required:
  - Technical Proposal
  - Past Performance Summary
  - Price Proposal

2 Technical Proposal Submission Requirements

The Technical Proposal must address all of the following:

2.1 Technical Approach

Offerors must describe in detail how they will perform:

- Trimming, cassette printing, processing, paraffin embedding (16 blocks × 58 samples) 1
- H&E staining (16 blocks × 58 samples)

- Immunostaining of 30 slides per sample with all required antibodies
- Digital 40× slide scanning, image storage, and access requirements

Proposal must include workflow descriptions, QC measures, and throughput capabilities.

2.2 Turnaround Time & Logistics

Offeror shall explain how it will meet:

- 5–10 business day turnaround after receiving samples
- Pickup/delivery logistics to the NIH main campus in Bethesda

2.3 Digital Platform Description

Offeror must document:

- Storage capacity (≥67 TB)
- User access controls for at least 3 Government personnel
- System capabilities: annotation, snapshots, raw file download
- Data retention for 12 months

2.4 Deliverables Plan

Offeror shall detail procedures to provide all required deliverables:

- All processed tissue blocks
- All stained slides
- All digital images and platform access
- One-time processing report

2.5 Staffing & Key Personnel

Proposal must include:

- Names & qualifications of personnel performing the work
- Identification of the Project Manager required by the SOW

2.6 Facilities & Equipment

Offeror must describe:

- Facilities used for processing, staining, and imaging
- Equipment used to meet required throughput
- Data storage and IT security measures

3. Past Performance Submission

Offerors shall provide:

- Up to three examples of similar neuropathology, histology, or digital imaging projects.
- Points of contact for verification.

(For LPTA, past performance is generally Pass/Fail: no negative relevant history.)

4. Price Proposal Submission

Offeror must:

- Provide a complete price breakdown (labor, materials, scanning, storage, slide prep, pickup/delivery).
- Ensure the price matches the work described.
- Submit in a separate file/section.

Price must reflect the full SOW workload of 58 brain samples and associated deliverables.

5. General Submission Instructions

- Late proposals will not be accepted.
- Offerors must follow all instructions exactly.
- The Government may reject any proposal that lacks required information.